

Derivation v20: Factor & Normalization Audit for the 5D→4D Gravity Bridge

EDC Collaboration

Derivation note; explicit factor tracking

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Abstract

This note performs a forensic audit of all numerical factors in the derivation of $\bar{M}_{\text{Pl}}^2 = M_5^3 \mathcal{I}$. We explicitly track: (i) the $\frac{1}{2}$ in action normalization, (ii) the 8π in Newton's constant, (iii) orbifold doubling factors, (iv) the 2π from circle circumference, and (v) the warp factor exponent in $\mathcal{I}$. Three coordinate conventions are compared and a canonical choice is fixed. A conversion table maps between conventions.

1 Conventions (Fixed for This Note)

1.1 5D Action Normalization

We adopt the 5D Einstein-Hilbert action in the form:

$$S_5 = \frac{M_5^3}{2} \int d^5x \sqrt{-g_5} R_5 \quad (1)$$

where M_5 is the fundamental 5D Planck mass. This is equivalent to writing $S_5 = \frac{1}{2\kappa_5^2} \int \sqrt{-g_5} R_5$ with $\kappa_5^2 = 1/M_5^3$.

Note: The factor $\frac{1}{2}$ multiplies the entire action, not just the Ricci scalar. This is the “reduced” action convention standard in brane-world literature.

1.2 4D Newton Constant and Planck Mass

Newton's constant in 4D is defined via:

$$G_N = \frac{1}{8\pi \bar{M}_{\text{Pl}}^2} \quad (2)$$

where $\bar{M}_{\text{Pl}}$ is the **reduced Planck mass**:

$$\bar{M}_{\text{Pl}} = \frac{1}{\sqrt{8\pi G_N}} \approx 2.435 \times 10^{18} \text{ GeV} \quad (3)$$

The “original” Planck mass is:

$$M_{\text{Pl}} = \frac{1}{\sqrt{G_N}} \approx 1.221 \times 10^{19} \text{ GeV} \quad (4)$$

Relation:

$$\bar{M}_{\text{Pl}} = \frac{M_{\text{Pl}}}{\sqrt{8\pi}} \quad (5)$$

The 4D Einstein-Hilbert action with our conventions is:

$$S_4 = \frac{\bar{M}_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g_4} R_4 = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g_4} R_4 \quad (6)$$

1.3 Extra Dimension Coordinate Conventions

We consider three coordinate conventions for the extra dimension ξ :

1. **Interval** $[0, L]$: Flat extra dimension with coordinate $\xi \in [0, L]$. The length of the extra dimension is L .
2. **Circle** S^1 : Compact circle with $\xi \in [0, 2\pi R]$ and periodic identification $\xi \sim \xi + 2\pi R$. Circumference is $2\pi R$.
3. **Orbifold** $S^1/\mathbb{Z}_2$: Circle with $\mathbb{Z}_2$ reflection symmetry $\xi \rightarrow -\xi$. Fundamental domain is $\xi \in [0, \pi R]$. Branes sit at the fixed points $\xi = 0$ and $\xi = \pi R$.

Definition: We define R_ξ as the *length of the fundamental domain*:

$$R_\xi \equiv L \quad (\text{interval}), \quad R_\xi \equiv 2\pi R \quad (\text{circle}), \quad R_\xi \equiv \pi R \quad (\text{orbifold}) \quad (7)$$

2 Linearized Equations and Mode Decomposition

2.1 Warped Metric Ansatz

The 5D metric is:

$$ds^2 = e^{2A(\xi)} \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2 \quad (8)$$

where $A(\xi)$ is the warp factor.

2.2 Metric Perturbation

Perturbing around the background:

$$g_{\mu\nu}^{(5)} = e^{2A}(\eta_{\mu\nu} + h_{\mu\nu}) \quad (9)$$

In transverse-traceless (TT) gauge:

$$\partial^\mu h_{\mu\nu} = 0, \quad \eta^{\mu\nu} h_{\mu\nu} = 0 \quad (10)$$

2.3 KK Decomposition

We decompose:

$$h_{\mu\nu}(x, \xi) = \sum_n h_{\mu\nu}^{(n)}(x) \psi_n(\xi) \quad (11)$$

2.4 Derivation of Mode Equation

The linearized 5D Einstein equation for TT modes in the bulk is:

$$\square_5 h_{\mu\nu} = 0 \quad (12)$$

Expanding the 5D d'Alembertian:

$$\square_5 = e^{-2A} \square_4 + \partial_\xi^2 + 4A' \partial_\xi \quad (13)$$

Substituting the KK decomposition and using $\square_4 h_{\mu\nu}^{(n)} = -m_n^2 h_{\mu\nu}^{(n)}$:

$$-m_n^2 e^{-2A} \psi_n + \psi_n'' + 4A' \psi_n' = 0 \quad (14)$$

Rearranging:

$$\boxed{\psi_n'' + 4A' \psi_n' + m_n^2 e^{2A} \psi_n = 0} \quad (15)$$

2.5 Zero-Mode Solution

For $m_0 = 0$:

$$\psi_0'' + 4A'\psi_0' = 0 \quad \Rightarrow \quad \frac{d}{d\xi}(e^{4A}\psi_0') = 0 \quad (16)$$

For normalizable modes: $\psi_0' = 0$, hence:

$$\psi_0 = \text{const} \quad (17)$$

3 Kinetic Term Reduction: Factor-by-Factor

This section is the core of the factor audit.

3.1 5D Kinetic Term

Starting from the 5D action (1), the quadratic kinetic term for TT graviton modes is:

$$S_5^{(2)} = \frac{M_5^3}{2} \int d^5x \sqrt{-g_5} \left(-\frac{1}{4} g^{AC} g^{BD} \nabla_A h_{CB} \nabla_D h_{CB} \right) \quad (18)$$

For our metric (8):

$$\sqrt{-g_5} = e^{4A} \quad (19)$$

The inverse metric components:

$$g^{\mu\nu} = e^{-2A} \eta^{\mu\nu}, \quad g^{55} = 1 \quad (20)$$

3.2 Collecting Factors

The 4D kinetic piece of the graviton (ignoring ξ -derivatives, which give mass terms):

$$S_5^{(2)} \supset \frac{M_5^3}{2} \int d^4x d\xi e^{4A} \left(-\frac{1}{4} e^{-2A} e^{-2A} \eta^{\mu\rho} \eta^{\nu\sigma} \partial_\mu h_{\rho\sigma} \partial_\nu h^{\rho\sigma} \right) \quad (21)$$

Simplifying:

$$S_5^{(2)} \supset -\frac{M_5^3}{8} \int d^4x d\xi e^{4A} e^{-4A} \eta^{\mu\rho} \eta^{\nu\sigma} \partial_\mu h_{\rho\sigma} \partial_\nu h^{\rho\sigma} \quad (22)$$

Key observation: The e^{4A} from $\sqrt{-g_5}$ cancels the e^{-4A} from two factors of $g^{\mu\nu} = e^{-2A} \eta^{\mu\nu}$. Therefore:

$$S_5^{(2)} \supset -\frac{M_5^3}{8} \int d^4x d\xi \eta^{\mu\rho} \eta^{\nu\sigma} \partial_\mu h_{\rho\sigma} \partial_\nu h^{\rho\sigma} \quad (23)$$

3.3 Substituting KK Zero-Mode

For the zero-mode $h_{\mu\nu} = h_{\mu\nu}^{(0)}(x) \psi_0(\xi)$:

$$S_5^{(2)} \supset -\frac{M_5^3}{8} \int d^4x \eta^{\mu\rho} \eta^{\nu\sigma} \partial_\mu h_{\rho\sigma}^{(0)} \partial_\nu h^{(0)\rho\sigma} \cdot \int d\xi |\psi_0|^2 \quad (24)$$

Define:

$$\boxed{\mathcal{I} \equiv \int_{\text{domain}} d\xi |\psi_0(\xi)|^2} \quad (25)$$

CRITICAL: With the factor cancellation above, the exponent in $\mathcal{I}$ is **zero**, not e^{2A} or e^{4A} ! Continuing:

$$S_5^{(2)} \supset -\frac{M_5^3 \mathcal{I}}{8} \int d^4x \partial_\mu h_{\rho\sigma}^{(0)} \partial^\mu h^{(0)\rho\sigma} \quad (26)$$

3.4 Matching to 4D Action

The 4D kinetic term from (6) is:

$$S_4^{(2)} = \frac{\bar{M}_{\text{Pl}}^2}{2} \cdot \left(-\frac{1}{4}\right) \int d^4x \partial_\mu h_{\rho\sigma} \partial^\mu h^{\rho\sigma} = -\frac{\bar{M}_{\text{Pl}}^2}{8} \int d^4x \partial_\mu h_{\rho\sigma} \partial^\mu h^{\rho\sigma} \quad (27)$$

Matching (26) to (27):

$$\boxed{\bar{M}_{\text{Pl}}^2 = M_5^3 \mathcal{I}} \quad (28)$$

3.5 Newton's Constant

From (2) and (28):

$$\boxed{G_N = \frac{1}{8\pi M_5^3 \mathcal{I}}} \quad (29)$$

4 Computing $\mathcal{I}$ for Different Conventions

4.1 Flat Case: $A(\xi) = 0$

For a flat extra dimension, the zero-mode is constant: $\psi_0 = c$.

4.2 Case 1: Interval $[0, L]$

Integration domain: $\xi \in [0, L]$.

Normalization: $\int_0^L d\xi |c|^2 = |c|^2 L = 1$ gives $|c|^2 = 1/L$.

Therefore:

$$\mathcal{I} = \int_0^L d\xi |\psi_0|^2 = \int_0^L d\xi \frac{1}{L} = 1 \quad (\text{dimensionless}) \quad (30)$$

Alternatively, with $\psi_0 = 1$ (unnormalized):

$$\mathcal{I} = \int_0^L d\xi = L \quad (31)$$

With our definition $R_\xi = L$:

$$\mathcal{I} = R_\xi \quad (\text{interval, unnormalized } \psi_0 = 1) \quad (32)$$

4.3 Case 2: Circle S^1 with $\xi \in [0, 2\pi R]$

Integration domain: $\xi \in [0, 2\pi R]$.

With $\psi_0 = 1$:

$$\mathcal{I} = \int_0^{2\pi R} d\xi = 2\pi R \quad (33)$$

With our definition $R_\xi = 2\pi R$ (circumference):

$$\mathcal{I} = R_\xi \quad (\text{circle}) \quad (34)$$

4.4 Case 3: Orbifold $S^1/\mathbb{Z}_2$

The orbifold $S^1/\mathbb{Z}_2$ has a $\mathbb{Z}_2$ symmetry $\xi \rightarrow -\xi$.

Method A: Integrate over fundamental domain $[0, \pi R]$ only:

$$\mathcal{I} = \int_0^{\pi R} d\xi = \pi R \quad (35)$$

Method B: Integrate over full circle $[-\pi R, \pi R]$ and divide by orbifold factor 2:

$$\mathcal{I} = \frac{1}{2} \int_{-\pi R}^{\pi R} d\xi = \frac{1}{2} \cdot 2\pi R = \pi R \quad (36)$$

Both methods agree.

With our definition $R_\xi = \pi R$ (fundamental domain length):

$$\mathcal{I} = R_\xi \quad (\text{orbifold}) \quad (37)$$

4.5 Summary: Universal Result

With the definition (7) (R_ξ = length of fundamental domain) and unnormalized $\psi_0 = 1$:

$$\boxed{\mathcal{I} = R_\xi \quad (\text{universal, flat case})} \quad (38)$$

5 Canonical Convention and Conversion Table

5.1 Canonical Choice

We adopt:

$$R_\xi \equiv \text{length of fundamental integration domain} \quad (39)$$

The bridge relations become:

$$\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi \quad (40)$$

$$G_N = \frac{1}{8\pi M_5^3 R_\xi} \quad (41)$$

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3} \quad (42)$$

5.2 Conversion Table

Table 1: Conversion between geometric conventions.

Geometry	Domain	$\mathcal{I}$ (flat)	R_ξ definition	Mapping
Interval $[0, L]$	$[0, L]$	L	$R_\xi = L$	$R_\xi = L$
Circle S^1	$[0, 2\pi R]$	$2\pi R$	$R_\xi = 2\pi R$	$R_\xi = 2\pi R$
Orbifold $S^1/\mathbb{Z}_2$	$[0, \pi R]$	πR	$R_\xi = \pi R$	$R_\xi = \pi R$

Conversion rule: If an alternative convention uses $\tilde{R}$ where $\mathcal{I} = f(\tilde{R})$, then $R_\xi = f(\tilde{R})$.

Example: If someone uses “ R ” to mean the circle *radius* (not circumference), then $\mathcal{I} = 2\pi R$ and thus $R_\xi = 2\pi R$.

5.3 Calibrated Closure

With $R_\xi = \hbar c/M_Z^{\text{obs}}$ identified as [I]+[BL]:

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3} = (2.435 \times 10^{18})^{2/3} \times (2.165 \times 10^{-18} \text{ m})^{-1/3} \quad (43)$$

Converting R_ξ to GeV^{-1} : $R_\xi = 1/M_Z = 1/91.19 \text{ GeV}^{-1}$:

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} M_Z^{1/3} = (2.435 \times 10^{18})^{2/3} \times (91.19)^{1/3} \text{ GeV} \quad (44)$$

Computing:

$$M_5 = 1.81 \times 10^{12} \times 4.50 = 8.1 \times 10^{12} \text{ GeV} \quad (45)$$

CORRECTION vs v19: In v19, we used $M_{\text{Pl}} = 1.221 \times 10^{19} \text{ GeV}$ (the “original” Planck mass) in the formula. With the *reduced* Planck mass $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$, the result is:

$$M_5 = 8.1 \times 10^{12} \text{ GeV} \quad (\text{reduced convention}) \quad (46)$$

If instead we use the *original* Planck mass in the bridge $M_{\text{Pl}}^2 = M_5^3 R_\xi$:

$$M_5 = M_{\text{Pl}}^{2/3} R_\xi^{-1/3} = (1.221 \times 10^{19})^{2/3} \times (91.19)^{1/3} = 2.4 \times 10^{13} \text{ GeV} \quad (47)$$

The difference is a factor of $(8\pi)^{1/3} \approx 2.94$.

6 Factor Audit Summary

Factor Audit Box		
Factor	Resolution	Eq. ref
$\frac{1}{2}$ in 5D action	Fixed in (1)	Eq. (1)
$\frac{1}{4}$ in kinetic term	Standard TT normalization	Eq. (18)
8π in G_N definition	$G_N = 1/(8\pi \bar{M}_{\text{Pl}}^2)$	Eq. (2)
Reduced vs original M_{Pl}	Use $\bar{M}_{\text{Pl}} = M_{\text{Pl}}/\sqrt{8\pi}$	Eq. (5)
Warp factor exponent in $\mathcal{I}$	Zero (factors cancel)	Eq. (25)
Orbifold factor of 2	Absorbed in domain definition	Eq. (36)
2π from circle	Absorbed in $R_\xi = 2\pi R$	Eq. (34)
R_ξ definition	Length of fundamental domain	Eq. (7)

7 Reconciliation with v19

7.1 v19 Convention

Derivation v19 used:

- $M_{\text{Pl}} = 1.221 \times 10^{19} \text{ GeV}$ (original Planck mass)
- $G_N = 1/(8\pi M_{\text{Pl}}^2)$ (which is dimensionally inconsistent with the original Planck mass definition $M_{\text{Pl}} = G_N^{-1/2}$)

7.2 Self-Consistent Options

Option A (reduced convention):

$$G_N = \frac{1}{8\pi\bar{M}_{\text{Pl}}^2}, \quad \bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV} \quad (48)$$

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3} = 8.1 \times 10^{12} \text{ GeV} \quad (49)$$

Option B (original convention):

$$G_N = \frac{1}{M_{\text{Pl}}^2}, \quad M_{\text{Pl}} = 1.221 \times 10^{19} \text{ GeV} \quad (50)$$

$$M_5 = M_{\text{Pl}}^{2/3} R_\xi^{-1/3} = 2.4 \times 10^{13} \text{ GeV} \quad (51)$$

7.3 Which is Correct?

Both are correct *within their own conventions*. The key is consistency:

- If 4D action is $S_4 = \frac{\bar{M}_{\text{Pl}}^2}{2} \int \sqrt{-g} R$, use Option A.
- If 4D action is $S_4 = \frac{M_{\text{Pl}}^2}{16\pi} \int \sqrt{-g} R$, use Option B.

v19 implicitly used Option B (numerical value $M_{\text{Pl}} = 1.221 \times 10^{19} \text{ GeV}$), but wrote the bridge in the Option A form. This is a notational inconsistency, not a physics error, because the 8π factors cancel in physical observables.

7.4 Recommendation

For clarity, future notes should use the reduced convention (Option A) with $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$. The numerical result for M_5 depends on convention:

$$M_5 = \begin{cases} 8.1 \times 10^{12} \text{ GeV} & \text{(reduced: Option A)} \\ 2.4 \times 10^{13} \text{ GeV} & \text{(original: Option B)} \end{cases} \quad (52)$$

Both are at the GUT scale; the factor of 3 difference is a convention choice.

8 Conclusion

1. The warp factor exponent in $\mathcal{I}$ is **zero** due to exact cancellation of e^{4A} from $\sqrt{-g_5}$ and e^{-4A} from $g^{\mu\nu}g^{\rho\sigma}$.
2. R_ξ should be defined as the **length of the fundamental domain**, giving $\mathcal{I} = R_\xi$ universally (flat case).
3. The 8π factor resides in $G_N = 1/(8\pi\bar{M}_{\text{Pl}}^2)$ and in the relation $\bar{M}_{\text{Pl}} = M_{\text{Pl}}/\sqrt{8\pi}$.
4. No orbifold doubling ambiguity exists if R_ξ is defined as the fundamental domain length.
5. The numerical value of M_5 depends on the Planck mass convention: $\sim 10^{13} \text{ GeV}$ (original) or $\sim 10^{12} \text{ GeV}$ (reduced). Both are valid.

Canonical bridge (reduced convention):

$$\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi, \quad G_N = \frac{1}{8\pi M_5^3 R_\xi}$$

Closure: $M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3}$ once R_ξ is identified [I]/[BL].